

Interview Experiences

Date : 23/08/2021

Platform : Microsoft Teams

Interview Experience 1:

Round 1:

Technical Round : 65 minutes

Platform - No such platform for coding questions. Had to use an editor on my computer and share my screen the whole time.

Interviewer had 5 years experience with microsoft.

She started by giving a short introduction and asked me to do the same about myself.

She asked me about my favourite subjects. I told her about DBMS and Data Structures and Algorithms.

DBMS questions.

What additional constraints are specified for Relational DBMS?

Normal Forms (detailed explanation). What was the need to introduce BCNF?

What is Referential integrity? Foreign key? Primary Key?

Can the foreign key be present in the same table? Give an example.

Types of joins and join algorithms in DBMS(detailed explanation).

Data Structure question.

She asked me to implement a heap.

First I had to explain my approach for the extractmin, insert and deletemin functions.

She asked me to code it. I had to design the node structure, then write all the 2 functions discussed above.

She was satisfied with what I wrote and I didn't have to code the deletemin function(Lucky Escape).

Coding Question 1:

Given a function with 2 integers as formal parameters. Return the larger number without using the Relational operators.(Can only use == operator)

I gave her an approach using their difference and then summation.

At first, she was doubtful about my solution but then gave 3-4 inputs. Outputs were all correct and she seemed satisfied.

Coding Question 2:

Given a Binary Tree. Convert it into a doubly linked list using the same data structure.

Told her the approach of traversing inorder and generating a new node everytime towards the end of DLL. She asked me to optimize i.e. solve by manipulating the pointers in the tree itself.

I had to code from scratch on my code editor. I wrote the code but it was throwing a runtime error. I kept on explaining every line and she was satisfied with it but she told that I had to debug on my own.

She didn't even allow me to write some print statements to debug my code.

We were going overtime, so she said we have to stop here.

I was able to answer almost all the questions, except for the last part where I had to debug the code.

She asked me if I had any questions. I asked her about what exactly do they expect from a fresher and her experience with microsoft.

We thanked each other and left.

Round 2:

Technical Interview : 70 minutes

Platform : Codility

The interviewer was very friendly and we took the first 5 minutes to introduce ourselves.

He then shared a codility session link where the question along with all the functions and testing code was prewritten.

Coding Question 1:

Given two sorted linked lists. Merge them to form one sorted list.

<https://leetcode.com/problems/merge-two-sorted-lists/>

It was more a discussion than an interview. We discussed 4 iterative approaches with little changes(one with extra memory, other with less memory wastage, one to avoid redundant condition checks etc.)

Then he asked me to solve this question recursively.

Programming Question 1:

He shared another session link with a prewritten code snippet and asked me the output of it.

I found that the program was throwing an error as a function was returning the address of one of it's local variables.

Then he went into the details of it regarding the operating system.

He asked questions like-

What kind of error will that be?

Who stops this program from accessing the memory which is not allocated to it anymore?

How does the OS handle such errors?

What is the role of signals while handling these errors?

Where is the variable's memory allocated,stack or heap segment?

He then asked to modify the code to remove this error. Since we had talked about stack and heap segments just before this question, I took the hint from that and allocated the memory explicitly (using malloc) for the variable to be returned. I also told him another method would be to use extern storage class. He was satisfied by both of these.

He asked me if I had any questions. I asked him about his work and the current project that he was working on.

Round 3:

HR/Project Based Interview : 30 minutes

The interviewer looked very senior. But he was also very friendly. He took 2 minutes to talk to me and to calm me down.

I gave my introduction. He asked me why I didn't pursue my career in badminton rather than studying.

He started the discussion about my project which was ML based. He asked me my motivation behind choosing this project.

Then he asked me to explain in detail my whole project in complete layman terms from scratch. I took around 12-15 minutes to explain everything in detail. Meanwhile he also asked many cross-questions.

Then came the HR questions:

[Interviewer]

Tell me something about yourself that you are proud of.

Tell me a situation where you made a promise to someone and you just gave it everything to keep that promise.

Tell me a situation when your team or subordinates had some complaint about you. How did you deal with that situation?

What are your strengths?

He was quite appreciative and that helped me a lot.

He asked me if I had any questions. I asked him about his experience, as everything went online due to the pandemic and how he adapted to those changes.

We thanked each other and left.

General Tips:

Keep talking about whatever comes to your mind(especially during coding interviews). It not only keeps the interviewer involved but also helps you not to lose focus.

Writing clean and readable codes is a must.

Never assume anything. Ask the interviewer whatever you feel relevant.

Keep your energy high throughout the interview. Keep smiling. These positive vibes coming from you will always keep them interested in what you say.

Lean in, listen to the interviewer, show you interest when he/she is answering your questions.

Greet everyone at the beginning and end of an interview (with a smile).

VERDICT : SELECTED

Interview Experience 2

Round 1(45 -60 minutes.)

First he asked 'how are u 'and then interviewer introduced himself and asked me to introduce.

Describe about your favorite project?

What difficulties u faced during project?

Then asked a coding question.

<https://leetcode.com/problems/simplify-path/description/>

(no text editor is provided. We have to use our own editor and run.)

Do u have any questions?

Experience 2:(45-60 minutes)

First 5-10 minutes:

He asked 'how are you' introduced himself, and asked me to introduce myself.

He gave general instructions

Next 20-25 minutes:

He asked me to tell abt my project which I mentioned on my resume. Then he asked me some cross questions and finally asked how we can optimize this

Some discussion about optimization.

. Then he asked abt the second project on my resume very brief.

Next 20-25 minutes:

Then He gave a codility link and asked me to solve a question

<https://www.geeksforgeeks.org/merge-two-sorted-linked-lists/>

Any specific questions for me?

Verdict : Rejected

Interview experience-3

Date: 23/08/2021

Round 1: (60mins)

It was a coding round. He asked two problems to code and in which second one has two subdivisions. It all started with greeting each other. He was super chill and he said he has 7yrs of experience at Microsoft. And he asked me to introduce myself briefly, then we talked about our likes and dislikes, then hobbies. Now he moved to the question. 1st was FIND MAX PATH SUM IN BINARY TREE which is a leetcode problem under hard category if I remember correctly. I used the proper name for functions and he started asking questions about the name like "why I named it as pivot?" Etc. He continuously asked me questions while typing and the code ran correctly in the first go itself. Some questions from OS like paging, segmentation, virtual memory, pros and cons of a few synchronisation mechanisms and ended with semaphores.

1st subdivision of 2nd question was PASCAL TRIANGLE which is also on leetcode I suppose. I started with the normal iterative version and he asked me if I could do that with recursion. After thinking for a while, the HCF method clicked in my mind. He asked me to prove why that is correct. I did it using proof by contradiction (using combination). He was satisfied with that. While compiling the code, a few errors were there and we both debugged it and printed the triangle. Both the coding questions were done on codility.

The 2nd subdivision of the 2nd question had 2 basic questions like how to swap 2 integers without a temporary variable and how to store states of three devices in one variable. He didn't want me to code for this and asked me to write the logic on paper and send him via email. After talking casually for a while he asked me if I had any questions for him. I asked him about his experience and his work at Microsoft. We thanked each other and left. Overall the round was a little heavy for me and the interviewer was quite understanding. I enjoyed it a lot.

Round 2: (55mins)

This round started with introducing each other. He was a senior employee and he said he has 18yrs of experience at Microsoft. He was also chill. After talking for a while about our hobbies

and all, he started with a coding question: MERGE K-sorted list which is on leetcode . I did it using a typical method and he asked me if I could do it better. So I used a heap there and the code ran in the first go. He was questioning me continuously about the approach I used, checked the edge cases thoroughly and moved to the next question. The question was FIND LONGEST PALINDROMIC SUBSTRING. I started with the Brute force approach and when he asked me to optimise it I was unable to do so. So he gave a hint like this problem can be solved using DP. Then I somehow found the optimal substructure and coded it on codility. After debugging , the code ran correctly . I was asked a few questions in DBMS from indexing and some questions about time complexity and space complexity . He asked me about the approach (basic pigeon hole principle) I used to solve one question in the screening test. After talking casually for a while, we thanked each other and left.

SOME TIPS FROM MY EXPERIENCE:

Avoid showing them how nervous you are and try to connect dots of something you don't know at that time before saying " I don't know".

Keep interacting with the interviewer and don't be reluctant to ask your doubts or take their help while solving questions.

Stay calm and keep smiling. Meanwhile, be conscious about what you speak to them.!!!!!!

Interview Experience - 4

Round 1 (45-50 Mins)

First the interviewer introduced himself and then asked me to do the same.

Then he moved on directly to the coding question and shared with me a codility link.

Coding question 1: <https://leetcode.com/problems/spiral-matrix/>

Explained the approach and started coding. Was about to finish and then he said to move on to the next question.

Coding question 2: <https://leetcode.com/problems/search-in-rotated-sorted-array/>

Explained 3 approaches and told me to code out the optimal approach.

He asked me if I had any questions. Then he answered the question and left.

Round 2 (45-50 Mins)

First the interviewer introduced himself and then asked me to do the same.

Then he asked me if i had done any internship and then asked about the project from my resume. Told to keep it short, so explained in 2 mins.

Then he moved on to the coding question and told me to share the screen with the IDE of my choice.

<https://www.geeksforgeeks.org/number-of-occurrences-of-2-as-a-digit-in-numbers-from-0-to-n/>

I first explained the Brute force solution then wrote its code. He gave me various test cases and he seemed satisfied as code passed all test cases.

Then he specifically asked me to write a recursive approach. I tried 5-10 mins for finding any connection and how previous work can be reused. As he saw me struggling, he gave me a hint and then I understood the concept and wrote code.

He didn't let me finish though and told me that he was just interested in my approach.

He asked me if I had any questions. Then he answered the question and left.

Round 3 (15 mins)

This interviewer was the friendliest of all. He started the interview by telling me to ask the questions directly. I took his interview for about 5 mins :p.

He didn't even ask for my introduction and talked casually.

He then asked for my favourite subject. I told him Operating Systems.

Why is OS your favourite subject?

What tasks are done by OS?

What happens when I open any file?

How is memory assigned to the process?

Can other processes access memory assigned to a process ?

Can multiple threads access memory together?

Do I need to change any code in Windows OS if Seagate comes out with a new type of HDD?

Cross questions and various discussions.

Then he thanked me and left.

Verdict : Selected

Interview Experience - 5

Round 1 (50 minutes)

After greetings, the interviewer asked me to introduce myself.

Then he directly jumped into the coding question. He shared the codility platform's link and asked me to write a code to print the level order traversal of the tree:

<https://www.geeksforgeeks.org/level-order-tree-traversal/>. I needed to complete the print function only.

He asked me if I knew OOPs. I said yes, then he asked me to write a code to depict inheritance. He also asked me some questions about the access specifiers and their uses.

Then he looked into my resume. As I had mentioned C/C++ in my resume, he gave me a code in C (related to dynamic memory allocation using pointers) and asked me for the output and also . He then asked me how the sizeof operator works.

Then he asked me some questions related to the operating system, like how the OS allocates the memory for the program and how it accesses it efficiently. Then he asked me what TLB is, how it works, and where it is stored and we had some discussion about paging and page faults. He asked me if I knew design patterns, I replied with a no.

At last he asked me if I had any question for him, I asked him about how Microsoft assigns roles for newly joined candidates.

Round 2:(55-60 minutes)

She started with her introduction and asked me to introduce myself. We took around 5 minutes for introduction and then she started with the coding question.

Coding Questions:

Q1 : Given a set of cities, the traveler needs to visit all the cities and return back to the source. Each city is connected with highways. The constraint was that each city can be visited any number of times, but the highways can be visited at most once. She just wanted the approach for this question and which Data Structure and Algorithm would I use. I explained some approaches. Then she added another constraint and asked me if the edges were weighted, then what would be the approach. I again explained some approaches and she was satisfied with my answer.

Q2: <https://www.techiedelight.com/generate-list-of-possible-words-from-a-character-matrix/>. She first asked me to explain the approach. I started with the brute force approach and explained the TC and data structure required. She asked me to optimize it then I told her the approach using the Trie data structure. She was satisfied with the approach and asked me to write the code from scratch in any code editor. I couldn't finish the code within time, so she asked me to explain the code I have made so far. I explained the code.

At last, she asked me if I had any questions for her. So I asked a few questions about the working environment.

Round 3:(20-25 minutes)

Interviewer joined the meeting and without any introduction, he directly asked me if I had any questions for him. I asked him about what an usual day of an employee in this position would be like.

He then looked into my resume and asked me some questions about my interests and hobbies, and asked some HR questions like relocation and why you want to work in Microsoft.

Since I had written C/C++ in the resume, he asked me why I didn't write Java and Python. I answered that I am familiar with them, but for competitive programming, I prefer C++. He asked me the reason for my preference, and I told him that C++ is faster than Java and Python. So he counter-questioned why C++ is faster. I explained some points, but my explanation did not convince him.

Then he gave me the following code snippet:

```
void foo()  
{  
int a = 3;  
int *b = new int(20);  
}
```

He asked if the code snippet would compile successfully. If yes, then what would the 2nd line do. I got confused and told that it would produce a compile-time error, so he asked me what kind of error it would produce, and I couldn't answer this properly.

We thanked each other and left the meeting.

Verdict : Rejected

General Tips from my experience:

Be thorough with the basics.

Maintain energy level and confidence throughout the interview..

Make sure to have in-depth knowledge of whatever you have mentioned in your resume. You should have reasons for every single statement written in the resume.

Don't make any assumptions, ask follow-up questions before you start writing the code. Make sure to understand the question properly before approaching the question.

Interview Experience 6 :

Round 1 (70 minutes) :

Introduce yourself.

Topic I feel free to discuss, I replied Data Structure and Algorithms.

Clone A Binary Tree with Random Pointer, Clone a Binary Tree with Random Pointers -
GeeksforGeeks.

This was unknown for me at that situation, i delivered Brute and Hashing approach but couldn't optimal one. He asked me to Code anywhere and share my screen. He didn't ask for main, only function writhing required.

This took a long and while working on it he asked a few technical questions.

Difference between Pass by Reference and Pass by Pointer.

What is Greedy

Benefits of BFS and DFS over each other.

Round 2 (75 minutes) :

He joined 15 minutes late but was extremely friendly. After taking the introduction he asked nothing technical but gave a real life scenario that we had to work together in that mean time with no specific input output format and constraints change very dynamically.

The Problem : Users are supposed to choose a few operations like fan on, fan off", "light on", "light off", heater... and there be a way to represent the current state of the room, like an alexa or google home and we have to implement this. For sake of simplicity the commands are string and few devices in room and not an AI based system. This process will be executed infinitely.

My approach : we can assign weightage to each device and $\text{Current_state} = \text{Sum}(\text{weight_device_i} * \text{isON}(\text{device_i}))$. So assigning weightage be careful. Then I implemented several modules, "input_meaning_extraction", "drop_current_operation" (ex if fan on comes 10 times so for 2nd to 10th this command has no use).

And few cross questions about the strategy while working together, ex why using hashing even this can be done using "if-else blocks".

So, the main idea of this interview was to check how the candidate can work with others.

Lastly, a few discussions about our personal life. (I also thought this never happen in any interview)

Round 3 (35 minutes) :

Interviewer was an old person but friendly nature and told we are not going to discuss anything technical.

Tell me about Yourself.

10 min discussions on PG Mini Project and cross questions from there.

Team Work Experience in your life

Promises you have to keep

Hobbies (then i had to explain him what is anime and how it's different from cartoon)

Integrity of your life

Past failures you had and what you learnt from them

Any questions for me

Verdict : Selected

General Tips from my experience :

Interview is completely different from reading interview experiences and listening to others experience; it's much easy and simple when your are in the room.

Interview be friendly enough to make you feel comfortable every time; you just need be confident, have a little smile and make good communication, even share the roughs of your mind with the interviewer.

Be diplomatic but be yourself in the meet, there is no way you can fake the interviewer.

And obviously have the complete knowledge of what you are explaining.